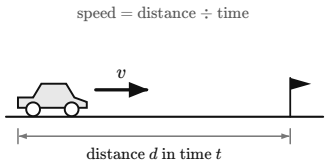
Formula Sheet

Grade 11 Physics · Speed, velocity and acceleration · Distance and displacement · Vectors · Coulomb's law

How to read it
- the equation
- the same equation rearranged, so every variable is the subject

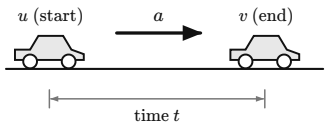
1 Speed, velocity and acceleration

speed · velocity · acceleration · equations of motion · motion graphs



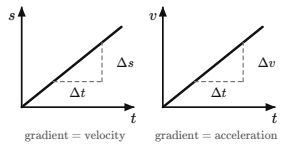
Distance, displacement and time

Average speed total distance ÷ total time	$v = \frac{d}{t}$	$d = vt$	$t = \frac{d}{v}$
Average velocity displacement ÷ time	$v = \frac{s}{t}$	$s = vt$	$t = \frac{s}{v}$
Journey in two stages add the distances, add the times	$v_{av} = \frac{d_1 + d_2}{t_1 + t_2}$	$d_2 = v_{av}(t_1 + t_2) - d_1$	$t_2 = \frac{d_1 + d_2}{v_{av}} - t_1$



Uniform acceleration from u to v

Acceleration rate of change of velocity	$a = \frac{v - u}{t}$	$v = u + at$	$u = v - at$	$t = \frac{v - u}{a}$
Displacement constant acceleration	$s = ut + \frac{1}{2}at^2$	$u = \frac{s - \frac{1}{2}at^2}{t}$	$a = \frac{2(s - ut)}{t^2}$	
Starting from rest u = 0	$s = \frac{1}{2}at^2$	$a = \frac{2s}{t^2}$	$t = \sqrt{\frac{2s}{a}}$	



Gradients of motion graphs

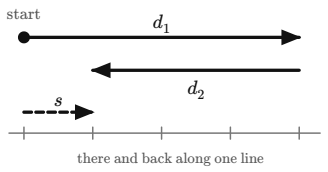
Gradient of s–t graph gives the velocity (d–t: speed)	$v = \frac{\Delta s}{\Delta t}$	$\Delta s = v\,\Delta t$	$\Delta t = \frac{\Delta s}{v}$
Gradient of v–t graph gives the acceleration	$a = \frac{\Delta v}{\Delta t}$	$\Delta v = a\,\Delta t$	$\Delta t = \frac{\Delta v}{a}$

Symbols v speed or velocity, m s^{-1} u initial velocity, m s^{-1} a acceleration, m s^{-2} d distance, m s displacement, m t time, s Δ change in

- Average speed is total distance ÷ total time, never the mean of the two speeds.
- Braking: a is negative (a deceleration). Flat line on an s–t graph: at rest; on a v–t graph: constant velocity.

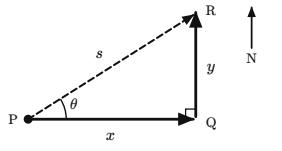
2 Distance and displacement

scalar and vector · journeys along a line · journeys at right angles



Opposite directions along one line

Total distance add every stage (scalar)	$d = d_1 + d_2$	$d_1 = d - d_2$	$d_2 = d - d_1$
Displacement opposite directions subtract	$s = d_1 - d_2$	$d_1 = s + d_2$	$d_2 = d_1 - s$
Back to the start one full lap or round trip	$s = 0$	average velocity $= \frac{0}{t} = 0$	



Two stages at right angles

Displacement at right angles Pythagoras	$s = \sqrt{x^2 + y^2}$	$x = \sqrt{s^2 - y^2}$	$y = \sqrt{s^2 - x^2}$	
Direction angle from the first stage	$\tan\theta = \frac{y}{x}$	$y = x\tan\theta$	$x = \frac{y}{\tan\theta}$	$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

Symbols d distance (scalar), m s displacement (vector), m x first stage (e.g. east), m y second stage (e.g. north), m θ angle of s from x

- Distance ≥ displacement; they are equal only for one straight stage in one direction.
- Always give a displacement with its direction, e.g. 1000 m at 53° north of east.

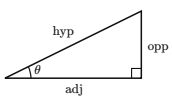
Constants

$k = 8.99 \times 10^9$ N m^2 C^{-2}
$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}
$e = 1.60 \times 10^{-19}$ C
$g = 9.81$ m s^{-2}

Prefixes and units

micro (μ) $= 10^{-6}$
nano (n) $= 10^{-9}$
1 h $= 3600$ s
km h^{-1} ÷ 3.6 = m s^{-1}
milli (m) $= 10^{-3}$
kilo (k) $= 10^3$
1 km $= 1000$ m

Right-angled triangle



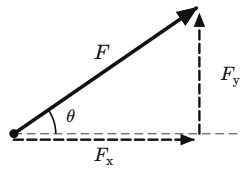
$\sin\theta = \frac{\text{opp}}{\text{hyp}}$ $\cos\theta = \frac{\text{adj}}{\text{hyp}}$ $\tan\theta = \frac{\text{opp}}{\text{adj}}$

$\text{hyp}^2 = \text{opp}^2 + \text{adj}^2$

Formula Sheet continued

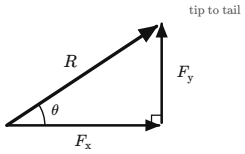
3 Vectors

resolving into components · combining components · resultant of two forces



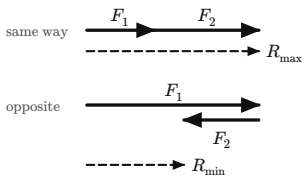
Resolving a vector into components

Horizontal component θ from the horizontal	$F_x = F\cos\theta$	$F = \frac{F_x}{\cos\theta}$	$\theta = \cos^{-1}\left(\frac{F_x}{F}\right)$
Vertical component θ from the horizontal	$F_y = F\sin\theta$	$F = \frac{F_y}{\sin\theta}$	$\theta = \sin^{-1}\left(\frac{F_y}{F}\right)$
θ from the vertical sin and cos swap	$F_x = F\sin\theta$	$F_y = F\cos\theta$	$F = \frac{F_x}{\sin\theta}$



Combining components into a resultant

Resultant Pythagoras	$R = \sqrt{F_x^2 + F_y^2}$	$F_x = \sqrt{R^2 - F_y^2}$	$F_y = \sqrt{R^2 - F_x^2}$
Direction angle from the horizontal	$\tan\theta = \frac{F_y}{F_x}$	$F_y = F_x\tan\theta$	$F_x = \frac{F_y}{\tan\theta}$	$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$



Largest and smallest resultant

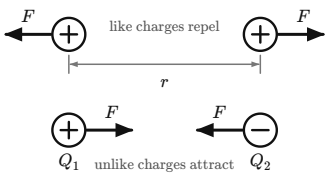
Same direction largest resultant	$R_{max} = F_1 + F_2$	$F_1 = R - F_2$	$F_2 = R - F_1$
Opposite directions smallest resultant	$R_{min} = F_1 - F_2$	$F_1 = R + F_2$	$F_2 = F_1 - R$
Any angle the resultant lies between	$F_1 - F_2 \le R \le F_1 + F_2$	e.g. 5 N and 12 N: $7\text{ N} \le R \le 17\text{ N}$	

Symbols F a force (any vector) F_x, F_y horizontal and vertical components R resultant F_1, F_2 two forces, F_1 the larger θ angle

- Check where θ is measured from before choosing sin or cos.
- The same rules work for velocity and displacement vectors.

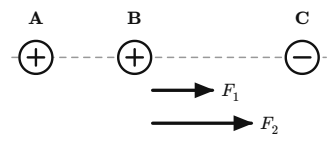
4 Coulomb's law

electrostatic force · two charges · three charges on a line · right-angled triangle



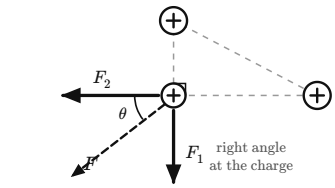
Force between two point charges

Coulomb's law any two point charges	$F = \frac{kQ_1Q_2}{r^2}$	$Q_1 = \frac{Fr^2}{kQ_2}$	$Q_2 = \frac{Fr^2}{kQ_1}$
	$r = \sqrt{\frac{kQ_1Q_2}{F}}$	$k = \frac{Fr^2}{Q_1Q_2}$	
Coulomb constant $k = 1/4\pi\varepsilon_0$	$k = \frac{1}{4\pi\varepsilon_0}$	$\varepsilon_0 = \frac{1}{4\pi k}$	
Changing Q or r inverse-square law	$F \propto \frac{Q_1Q_2}{r^2}$	$r \times 2 \Rightarrow F \div 4$	$Q_1 \times 2 \Rightarrow F \times 2$



A repels B and C attracts B: same way
Three charges on a straight line

Same line, same way forces add	$F = F_1 + F_2$	$F_1 = F - F_2$	$F_2 = F - F_1$
Same line, opposite ways forces subtract	$F = F_1 - F_2$	$F_1 = F + F_2$	$F_2 = F_1 - F$
Midpoint of equal charges equal and opposite	$F_1 = F_2 \Rightarrow F = 0$		



Charge at the right angle of a triangle

At right angles Pythagoras	$F = \sqrt{F_1^2 + F_2^2}$	$F_1 = \sqrt{F^2 - F_2^2}$	$F_2 = \sqrt{F^2 - F_1^2}$
Direction of the resultant angle from F_2	$\tan\theta = \frac{F_1}{F_2}$	$F_1 = F_2\tan\theta$	$F_2 = \frac{F_1}{\tan\theta}$	$\theta = \tan^{-1}\left(\frac{F_1}{F_2}\right)$
Two equal forces $F_1 = F_2$ at 90°	$F = \sqrt{2}\,F_1$	$F_1 = \frac{F}{\sqrt{2}}$	

Symbols F electrostatic force, N Q charge, C r distance between centres, m k Coulomb constant ε_0 permittivity of free space F_1, F_2 forces on one charge, N θ angle

- Convert μC to C ($\times 10^{-6}$) and cm to m before substituting.
- Like charges repel, unlike charges attract. Add forces on one charge as vectors: along a line add or subtract; at 90° use Pythagoras.